

Experiment 04

Aim

To implement Static Routing using multiple routers in Cisco Packet Tracer and verify communication between different networks.

Objective

1. To understand the concept of routing in computer networks.
 2. To study the working of static routing.
 3. To configure static routes on Cisco routers using Cisco IOS.
 4. To establish communication between different networks through multiple routers.
 5. To verify network connectivity using the ping command.
 6. To observe packet forwarding using Simulation Mode in Cisco Packet Tracer.
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Procedure

1. Open Cisco Packet Tracer and create a new workspace.
2. Place the required number of routers, switches, and PCs according to the network topology.
3. Connect the devices using the appropriate Ethernet and Serial cables.
4. Assign IP addresses and subnet masks to all PCs and router interfaces.
5. Configure the router interfaces using Cisco IOS commands and enable them using the no shutdown command.
6. Configure static routes on each router using the ip route command to define paths to remote networks.
7. Verify the routing configuration using commands such as show ip route and show ip interface brief.

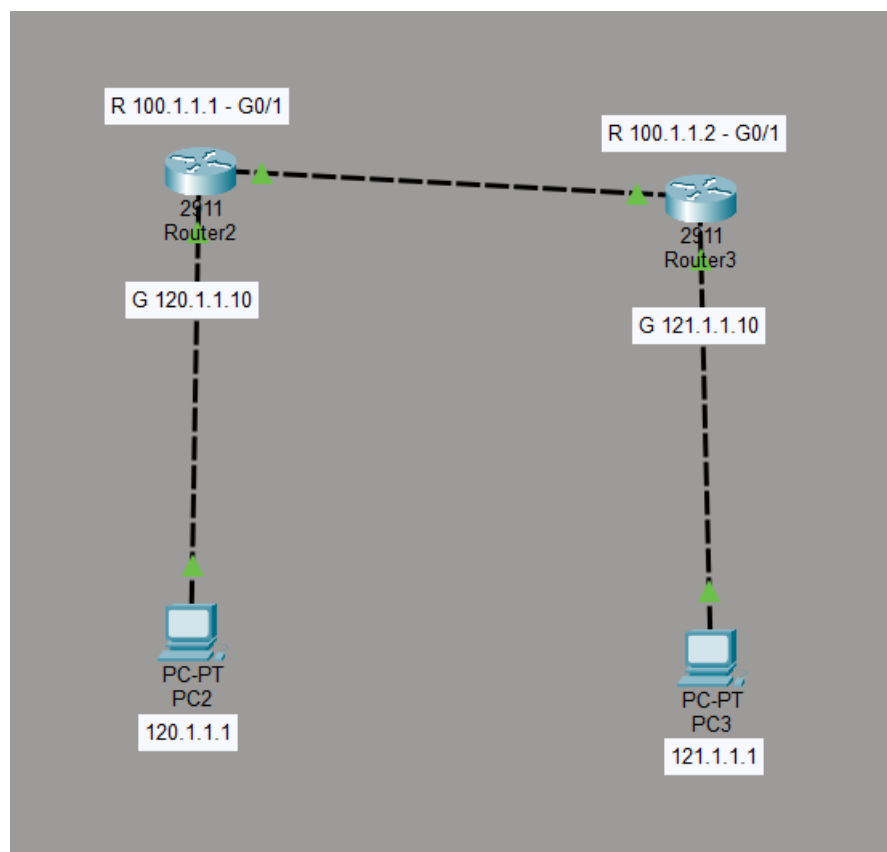
8. Test communication between PCs located on different networks using the ping command.
9. Observe packet transmission in Simulation Mode and verify successful routing.
10. Save the Packet Tracer project after successful implementation.

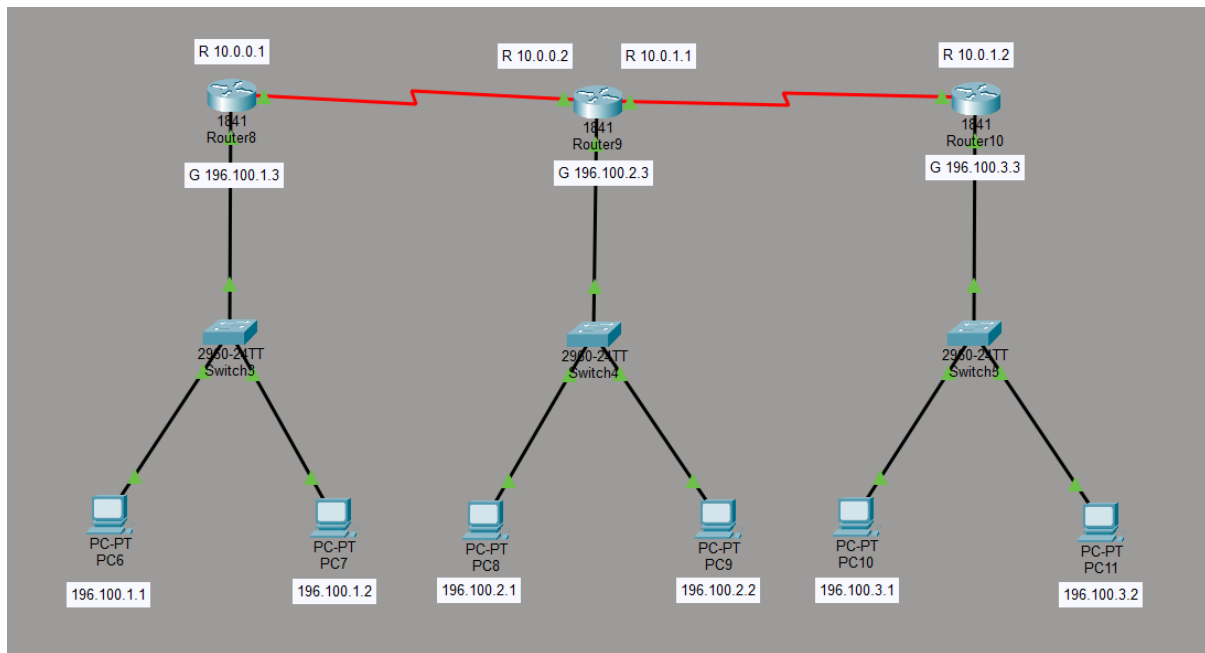
Input

Network topology consisting of multiple routers, switches, PCs, IP addresses, subnet masks, and static routing commands configured in Cisco Packet Tracer.

Output

Successful communication between devices located on different networks through manually configured static routes, verified using the ping command and routing table.





Learning Outcomes

After completing this experiment, the student was able to:

1. Understand the concept and importance of routing in computer networks.
2. Differentiate between static routing and dynamic routing.
3. Configure IP addresses and router interfaces in Cisco Packet Tracer.
4. Configure static routes using Cisco IOS commands.
5. Verify routing tables using the show ip route command.